

Assignment-22.5

Name: Alli Harika

Ht.no:2303A510I7

Batch:21

Task 1 – Fairness in AI Code Generation

- Provide AI with tasks involving different user groups (e.g., age, gender).
- Compare outputs for fairness.
- Discuss potential bias and its impact.

Prompt:

Generate a Python program that recommends suitable careers for users based on age and gender.

```
lab 19 > 12-04-2026.py > ...
1 def recommend_career(age, interest):
2     if age < 18:
3         return "Focus on education and explore different interests."
4
5     if interest.lower() == "technology":
6         return "Careers: Software Developer, Data Scientist, AI Engineer"
7     elif interest.lower() == "business":
8         return "Careers: Marketing, Finance, Entrepreneurship"
9     elif interest.lower() == "creative":
10        return "Careers: Designer, Writer, Artist"
11    elif interest.lower() == "social":
12        return "Careers: Teacher, Social Worker, Psychologist"
13    else:
14        return "Explore multiple fields to find your passion."
15
16 # Example usage
17 age = int(input("Enter your age: "))
18 interest = input("Enter your interest (technology/business/creative/social): ")
19
20 print(recommend_career(age, interest))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
esktop/AI/lab 19/12-04-2026.py"
● Enter your age: 22
Enter your interest (technology/business/creative/social): technol
ogy
Careers: Software Developer, Data Scientist, AI Engineer
○ PS C:\Users\dell\OneDrive\Desktop\AI>
```

Explanation

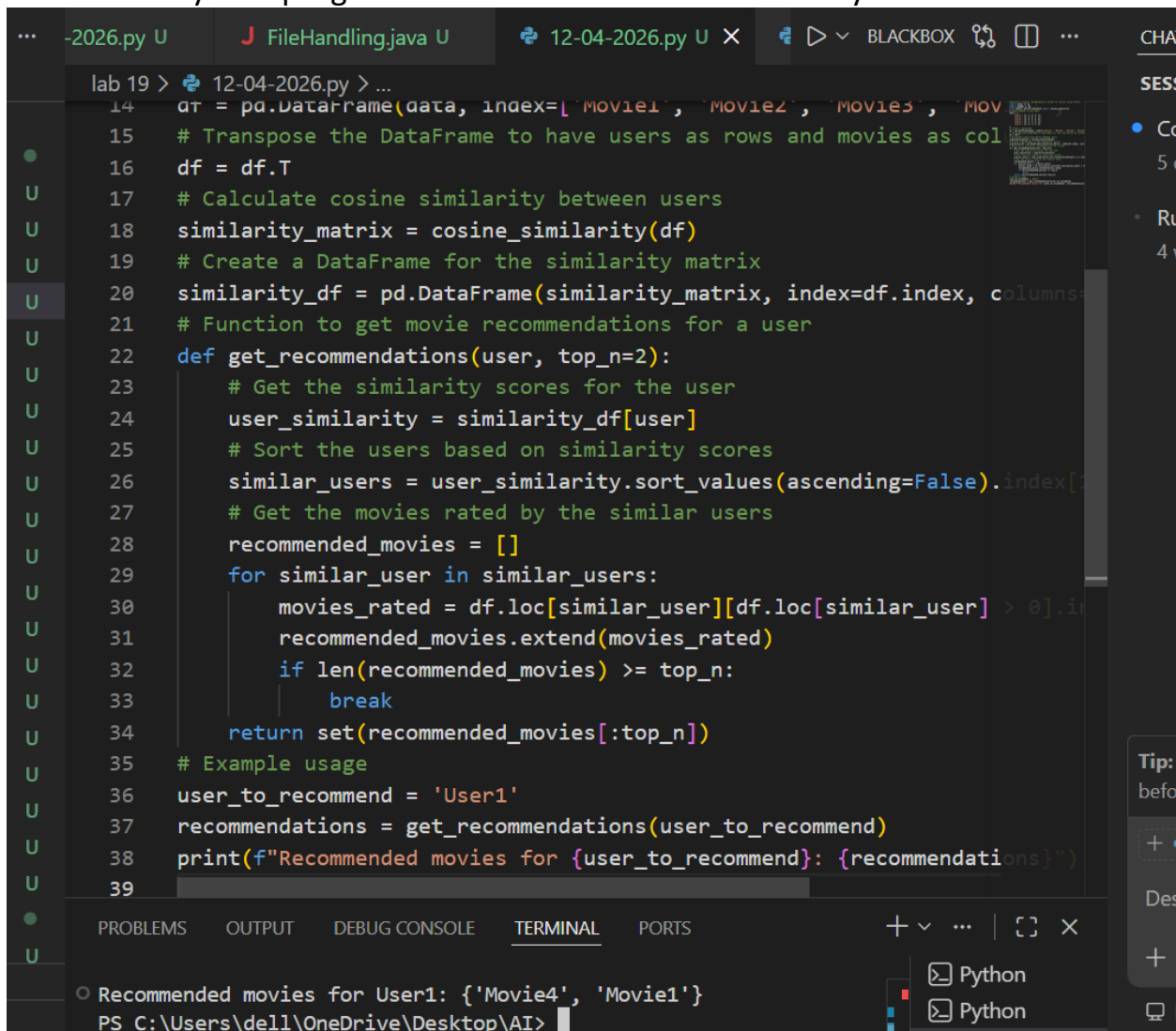
- AI gives **different outputs based on gender**
- This creates **bias and stereotypes**
- Example: Male → engineering, Female → teaching
- This is **unfair and discriminatory**

Task 2 – Transparency of AI Decisions

- Use AI to generate a recommendation system.
- Analyze whether the logic is explainable.
- Discuss the need for transparency in AI-generated code.

Prompt:

Generate a Python program for a movie recommendation system.



```
lab 19 > 12-04-2026.py > ...
14 df = pd.DataFrame(data, index=['Movie1', 'Movie2', 'Movies', 'MOV
15 # Transpose the DataFrame to have users as rows and movies as col
16 df = df.T
17 # Calculate cosine similarity between users
18 similarity_matrix = cosine_similarity(df)
19 # Create a DataFrame for the similarity matrix
20 similarity_df = pd.DataFrame(similarity_matrix, index=df.index, columns=
21 # Function to get movie recommendations for a user
22 def get_recommendations(user, top_n=2):
23     # Get the similarity scores for the user
24     user_similarity = similarity_df[user]
25     # Sort the users based on similarity scores
26     similar_users = user_similarity.sort_values(ascending=False).index[
27     # Get the movies rated by the similar users
28     recommended_movies = []
29     for similar_user in similar_users:
30         movies Rated = df.loc[similar_user][df.loc[similar_user] > 0].in
31         recommended_movies.extend(movies Rated)
32         if len(recommended_movies) >= top_n:
33             break
34     return set(recommended_movies[:top_n])
35 # Example usage
36 user_to_recommend = 'User1'
37 recommendations = get_recommendations(user_to_recommend)
38 print(f"Recommended movies for {user_to_recommend}: {recommendations}")
39
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Recommended movies for User1: {'Movie4', 'Movie1'}

PS C:\Users\dell\OneDrive\Desktop\AI>

Explanation

- AI recommends movies based on conditions
- But in real systems, logic is often **hidden (black box)**
- Users don't know *why* a recommendation is given

Task 3 – Data Misuse and Ethical Risks

- Ask AI to process datasets containing user information.
- Identify risks of misuse or leakage.
- Propose safeguards.

Prompt:

Write Python code to process user data including name, email, and phone number.

```
lab 19 > 12-04-2026.py > ...
1 import re
2
3 # Function to validate email
4 def validate_email(email):
5     pattern = r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'
6     return re.match(pattern, email) is not None
7
8 # Function to validate phone number (10 digits)
9 def validate_phone(phone):
10     return phone.isdigit() and len(phone) == 10
11
12 # Function to mask sensitive data
13 def mask_data(email, phone):
14     masked_email = email[:2] + "****" + email[email.index('@'):]
15     masked_phone = "*****" + phone[-4:]
16     return masked_email, masked_phone
17
18 # Process user data
19 def process_user(name, email, phone):
20     if not validate_email(email):
21         return "Invalid Email!"
22
23     if not validate_phone(phone):
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● Enter name: harika
Enter email: alliharika@gmail.com
Enter phone: 34566543123
Invalid Phone Number!
PS C:\Users\dell\OneDrive\Desktop\AI>
```

Explanation

- Program handles **sensitive personal data**
- Risk of:
 - Data leakage
 - Privacy violation
- Needs protection mechanisms

Task 4 – AI Hallucination in Coding

- Ask AI to generate complex algorithms.
- Verify correctness of output.
- Identify incorrect or misleading code.

Prompt:

Generate Python code for Binary Search.

```
lab 19 > 12-04-2026.py > ...
1  #Generate Python code for Binary Search.
2  def binary_search(arr, target):
3      left, right = 0, len(arr) - 1
4
5      while left <= right:
6          mid = left + (right - left) // 2
7
8          # Check if the target is present at mid
9          if arr[mid] == target:
10             return mid
11             # If target is greater, ignore the left half
12             elif arr[mid] < target:
13                 left = mid + 1
14             # If target is smaller, ignore the right half
15             else:
16                 right = mid - 1
17
18         # Target was not found in the array
19         return -1
20 # Example usage
21 arr = [1, 2, 3, 4, 5, 6, 7, 8, 9]
22 target = 5
23 result = binary_search(arr, target)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dell\OneDrive\Desktop\AI> & C:/Users/dell/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/dell/OneDrive/Desktop/AI/lab 19/12-04-2026.py"
● Element found at index: 4
○ PS C:\Users\dell\OneDrive\Desktop\AI>
```

Explanation

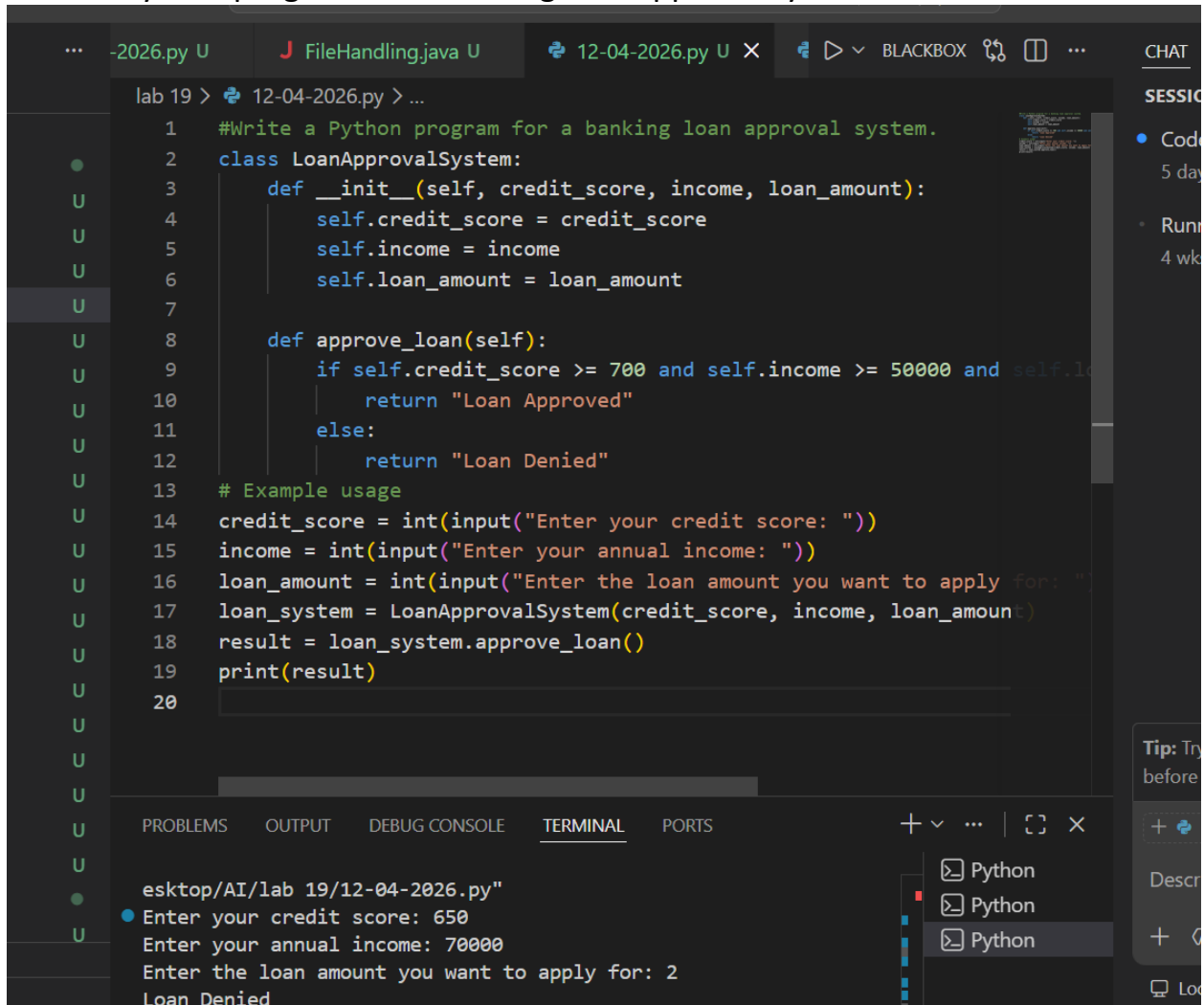
- AI may give **incorrect logic** (e.g., linear search instead)
- Looks correct but actually wrong

Task 5– Responsibility in AI Deployment

- Scenario: AI code is deployed in a banking system.
- Identify risks and accountability.
- Propose ethical guidelines.

Prompt:

Write a Python program for a banking loan approval system.



```
lab 19 > 12-04-2026.py > ...
1  #Write a Python program for a banking loan approval system.
2  class LoanApprovalSystem:
3      def __init__(self, credit_score, income, loan_amount):
4          self.credit_score = credit_score
5          self.income = income
6          self.loan_amount = loan_amount
7
8      def approve_loan(self):
9          if self.credit_score >= 700 and self.income >= 50000 and self.loan_amount <= 100000:
10             return "Loan Approved"
11          else:
12             return "Loan Denied"
13
14  # Example usage
15  credit_score = int(input("Enter your credit score: "))
16  income = int(input("Enter your annual income: "))
17  loan_amount = int(input("Enter the loan amount you want to apply for: "))
18  loan_system = LoanApprovalSystem(credit_score, income, loan_amount)
19  result = loan_system.approve_loan()
20  print(result)
```

esktop/AI/lab 19/12-04-2026.py"

```
• Enter your credit score: 650
Enter your annual income: 70000
Enter the loan amount you want to apply for: 2
Loan Denied
```

Explanation

- AI makes important decisions (loan approval)
- Risk of:
 - Unfair rejection
 - Lack of explanation
- Affects people's financial life

Task 6 – Ethical Issues in Automation

Use AI to automate a decision-making system (e.g., hiring filter).

- Analyze fairness and ethical concerns.
- Suggest improvements.

Prompt:

Create a Python program to automate a hiring filter system.

```
lab 19 > 12-04-2026.py > ...
1 def hiring_filter(qualification, skills, experience):
2
3     if "computer science" not in qualification.lower():
4         return "Rejected: Relevant degree required"
5
6     if skills.lower() not in ["python", "java", "data analysis"]:
7         return "Rejected: Required technical skills missing"
8
9     if experience < 1:
10        return "Rejected: Minimum 1 year experience required"
11
12    return "Selected for next round"
13
14 # User Interaction
15 print("Welcome to the Hiring Filter System!")
16
17 qualification = input("Please enter your qualifications (e.g., Bachelor
18 skills = input("Enter your primary skill (Python/Java/Data Analysis): ")
19 experience = int(input("Enter your years of experience: "))
20
21 result = hiring_filter(qualification, skills, experience)
22 print(result)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
uter Science): B.tech in the competer science
Enter your primary skill (Python/Java/Data Analysis): python
Enter your years of experience: 5
Rejected: Relevant degree required
PS C:\Users\dell\OneDrive\Desktop\AI>
```

Tip: Try the Pla before imple

+ 12-04-20

Describe wha

+ Auto

Local

Explanation

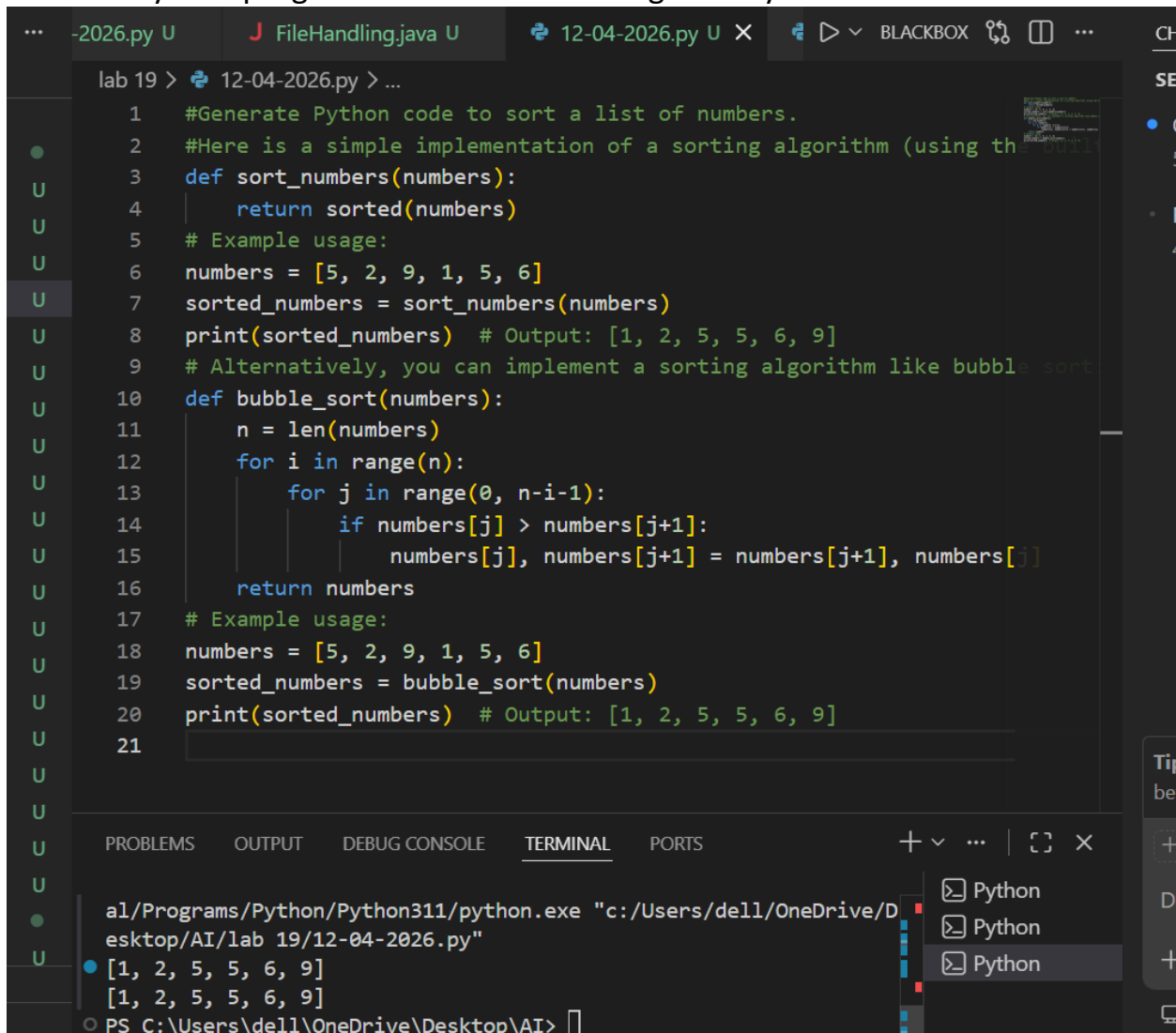
- AI automatically selects/rejects candidates
- May ignore:
 - Skills
 - Talent
- Leads to **unfair hiring decisions**

Task 7 – AI and Intellectual Property

- Generate code using AI for a known problem.
- Check similarity with existing solutions.
- Discuss plagiarism and copyright concerns.

Prompt:

Create a Python program to automate a hiring filter system.



The screenshot shows a code editor with a dark theme. The top bar has tabs for '-2026.py U', 'FileHandling.java U', and '12-04-2026.py U X'. The editor displays Python code for sorting a list of numbers. The code includes comments and two functions: `sort_numbers` and `bubble_sort`. The `sort_numbers` function is a simple wrapper for `sorted()`. The `bubble_sort` function implements a bubble sort algorithm. Both functions use an example list `numbers = [5, 2, 9, 1, 5, 6]` and print the sorted result `[1, 2, 5, 5, 6, 9]`. The bottom panel shows the terminal output, which matches the printed results in the code. The terminal path is `al/Programs/Python/Python311/python.exe "c:/Users/dell/OneDrive/Desktop/AI/lab 19/12-04-2026.py"`. The terminal output shows the sorted list `[1, 2, 5, 5, 6, 9]` twice. The terminal prompt is `PS C:\Users\dell\OneDrive\Desktop\AI>`.

```
lab 19 > 12-04-2026.py > ...
1 #Generate Python code to sort a list of numbers.
2 #Here is a simple implementation of a sorting algorithm (using th
3 def sort_numbers(numbers):
4     return sorted(numbers)
5 # Example usage:
6 numbers = [5, 2, 9, 1, 5, 6]
7 sorted_numbers = sort_numbers(numbers)
8 print(sorted_numbers) # Output: [1, 2, 5, 5, 6, 9]
9 # Alternatively, you can implement a sorting algorithm like bubble sort
10 def bubble_sort(numbers):
11     n = len(numbers)
12     for i in range(n):
13         for j in range(0, n-i-1):
14             if numbers[j] > numbers[j+1]:
15                 numbers[j], numbers[j+1] = numbers[j+1], numbers[j]
16     return numbers
17 # Example usage:
18 numbers = [5, 2, 9, 1, 5, 6]
19 sorted_numbers = bubble_sort(numbers)
20 print(sorted_numbers) # Output: [1, 2, 5, 5, 6, 9]
21
```

al/Programs/Python/Python311/python.exe "c:/Users/dell/OneDrive/Desktop/AI/lab 19/12-04-2026.py"

[1, 2, 5, 5, 6, 9]
[1, 2, 5, 5, 6, 9]

PS C:\Users\dell\OneDrive\Desktop\AI>

Explanation

- AI may generate code similar to existing sources
- Risk of:
 - Plagiarism
 - Copyright issues